

### 3.2.4 Density and pressure

| Learning outcomes                                                                             | Additional guidance |
|-----------------------------------------------------------------------------------------------|---------------------|
| <i>Learners should be able to demonstrate and apply their knowledge and understanding of:</i> |                     |
| (a) density; $\rho = \frac{m}{V}$                                                             | M0.1, M4.3          |
| (b) pressure; $p = \frac{F}{A}$ for solids, liquids and gases                                 |                     |
| (c) $p = h\rho g$ ; upthrust on an object in a fluid; Archimedes' principle.                  | M2.1<br>HSW4, 7, 11 |